Rationality of the center of a generic division algebra with a group action over a field of characteristic 0

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Abstract

Let F be a field of characteristic 0 and let $H = C_3 \times C_3$ act regularly as a transitive subgroup of S_9. We resolve Problem 5.2 of Auel–Brussel–Garibaldi–Vishne for stable rationality and rationality: $Z_H(F, 9)$, the center of the generic division algebra of degree 9 with H-action, is *not* stably rational, *not* rational, and *not* retract rational. The non-stable-rationality and the non-retract-rationality are proved by computing $\mathrm{H}^2(H, M|_H) \cong C_9$ for the restricted Procesi lattice and showing that its exponent 9 is incompatible with quasi-permutation and with every finite-index permutation superlattice, while all permutation lattices have H^2-exponent dividing 3. This integral cohomological obstruction is of a different nature from the classical unramified Brauer group. The proof generalizes to all odd primes p: for $H = C_p \times C_p \subset S_{p^2}$, one obtains $\mathrm{H}^2(H, M|_H) \cong C_{p^2}$ while permutation lattices have exponent dividing p, so $Z_H(F, p^2)$ is not stably rational.

Keywords: generic division algebra, stable rationality, integral group cohomology, permutation lattices.

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1 Introduction

Let F be a field of characteristic 0. For an integer $n \geq 1$ and a subgroup $H \subseteq S_n$, the *generic division algebra with H-action*, denoted $UD(F, n, H)$, has center $Z_H(F, n)$. Problem 5.2 of Auel–Brussel–Garibaldi–Vishne [1, Problem 5.2] asks:

> Determine if $Z_H(F, 9)$ is rational, stably rational, or retract rational over F, where $H = C_3 \times C_3 \subset S_9$ is transitive.

Let G be a finite group and let M be a $\mathbb{Z}[G]$-lattice, that is, a finitely generated $\mathbb{Z}$-free $\mathbb{Z}[G]$-module. Write $F(M)$ for the fraction field of the group algebra $F[M]$, that is, the function field of the torus with character lattice M, on which G acts through the $\mathbb{Z}[G]$-module structure of M; write $F(M)^G$ for its G-invariant subfield. For a subgroup H of G, write $M|_H$ for M regarded as a $\mathbb{Z}[H]$-lattice by restriction of scalars. Cyclic groups are denoted C_n, the augmentation ideal is $I[G] = \ker(\mathbb{Z}[G] \twoheadrightarrow \mathbb{Z})$, where the augmentation map sends $\sum n_g g \mapsto \sum n_g$, and $\exp(A)$ denotes the exponent of a finite abelian group A. Write $\mathrm{H}^i(G, M)$ for the i-th integral group cohomology of a G-module M; for $\mathbb{Z}[G]$-lattices this coincides with Tate cohomology in positive degrees [16, Def. 6.2.4].

A field extension K/F is *rational* if K is purely transcendental over F, and *stably rational* if $K(t_1, \ldots, t_m)$ is rational over F for some $m \geq 0$.

The rationality problem for fields of the form $F(M)^G$ has a long history. Endo and Miyata [7, §1] developed the theory of quasi-permutation modules and introduced the notion of quasi-rational extensions, providing an algebraic foundation for the rationality problem. Let G be a finite group, K/k a Galois extension with group G, and M a $\mathbb{Z}[G]$-lattice. Write $K(M)$ for the fraction field of the group algebra $K[M]$, on which G acts through the $\mathbb{Z}[G]$-module structure of M and the Galois action on K. They proved [8, Theorem 1.6] that $K(M)^G$ is *quasi-rational* over k if and only if M is a quasi-permutation $\mathbb{Z}[G]$-lattice. Voskresenskii [15, Chapter 2] gave a comprehensive treatment via the birational geometry of algebraic tori. Colliot-Thélène and Sansuc [5, §1] developed flasque resolutions and used them to characterize stable rationality of such fields in general by the vanishing of the flasque class. Saltman [13, Section 3] introduced retract rationality and constructed the first examples of non-rational stably rational fields.

When $H \subset S_n$ is cyclic and transitive, $Z_H(F, n)$ is rational if F contains a primitive n-th root of unity (and retract rational for general F when n is not of the form $8m$) [1, pp. 234]. Thus $H = C_3 \times C_3$ is the first case where the rationality question is genuinely open for n odd.

Property	*Status*	*Method*
Stably rational	**No (char 0)**	$\mathrm{H}^2(H, M\|_H) \cong C_9$
Rational	**No (char 0)**	rational $\Rightarrow$ stably rational
Retract rational	**No (char 0)**	Theorem 4.11

Table 1: Status of the rationality properties of $Z_H(F, 9)$ for $H = C_3 \times C_3$.

In this paper we determine the stable rationality and rationality of $Z_H(F, 9)$:

The main new ingredient is the computation of the second integral cohomology group of the restricted Procesi lattice $M|_H$: $\mathrm{H}^2(H, M|_H) \cong C_9$ (Lemma 4.2). The exponent 9 of this group directly obstructs quasi-permutation: we prove that for any permutation $\mathbb{Z}[H]$-lattice P (Definition 2.1), every element of $\mathrm{H}^2(H, P)$ has order dividing 3 (Proposition 4.4). Consequently $M|_H$ cannot be quasi-permutation (Lemma 4.7), and by Theorem 2.7, $Z_H(F, 9)$ is not stably rational over F (Corollary 4.8). This integral cohomological obstruction appears to be new for *multiplicative invariant fields*, i.e. the fields $F(M)^G$ above.

The logical chain is:

$$
\begin{aligned}
&\mathrm{H}^2(H, M|_H) \cong C_9 \\
&\quad\Longrightarrow\ M|_H \text{ is not quasi-permutation} \qquad \text{(Lemma 4.7)} \\
&\quad\Longrightarrow\ Z_H(F, 9) \text{ is not stably rational} \qquad \text{(Theorem 2.7, assumed)} \\
&\quad\Longrightarrow\ Z_H(F, 9) \text{ is not rational} \qquad \text{(rationality } \Rightarrow \text{ stable rationality).}
\end{aligned}
$$

Thus both non-stable-rationality and non-rationality hold over any field of characteristic 0 (via Theorem 2.7, used as a hypothesis), without reliance on its converse. The proof generalizes verbatim to all odd primes p (Theorem 5.1).

Hypotheses. The original statement in [1, Problem 5.2] is made over an arbitrary field. The cohomological computations ($\mathrm{H}^2(H, M|_H) \cong C_9$ and $\exp(\mathrm{H}^2(H, P)) \mid 3$) are purely integral, valid whenever $\operatorname{char} F \nmid 9$. The geometric conclusions (non-stable-rationality, non-rationality, non-retract-rationality) require $\operatorname{char} F = 0$ (the criteria are stated in characteristic 0): Theorem 2.7 and Theorem 4.10, used as hypotheses. None of the three conclusions relies on the converse implication of Theorem 2.7.

Notation. For the main case (Problem 5.2), $H = C_3 \times C_3 = \langle a, b \mid a^3 = b^3 = 1,\ ab = ba \rangle$; the generalization to $H = C_p \times C_p$ is treated in §5. The base field F is of characteristic 0. For a finite group G, a G-lattice is a finitely generated $\mathbb{Z}[G]$-module that is $\mathbb{Z}$-free. The group ring is denoted $\mathbb{Z}[G]$.

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Use of Large Language Models, AI and Machine Learning Tools. Claude Code and DeepSeek Harness, both with DeepSeek V4.1 Flash as the underlying LLM, were used under the author's direction and review, for literature searching, language polishing, debugging GAP scripts, the verification programs in the accompanying code package, and the machine-checked Lean development; in its present state that development is conditional on the two named hypotheses of the Hypotheses paragraph, the Endo–Miyata criterion (Theorem 2.7) and the retract-rationality criterion (Theorem 4.10).

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Data availability. The accompanying code package, available at `https://github.com/wuzhengyao/abgv52-lean`, contains the machine-checked Lean development and the two independent verification programs for the cohomological readings used in this paper, written in Python and in GAP using HAP, each with built-in self-tests; it also records their output for $p = 3$ and $p = 5$ and a `SHA-256` manifest.

2 Preliminaries

Definition 2.1. A $\mathbb{Z}[G]$-*lattice* is a finitely generated $\mathbb{Z}[G]$-module that is $\mathbb{Z}$-free. A $\mathbb{Z}[G]$-lattice M is called:

1. *permutation* if $M \cong \bigoplus_{i=1}^{r} \mathbb{Z}[G/K_i]$ for subgroups $K_i \leq G$, where $\mathbb{Z}[G/K]$ denotes the $\mathbb{Z}[G]$-module with $\mathbb{Z}$-basis the left cosets G/K and G-action by left multiplication;

2. *stably permutation* if there exist permutation lattices P, Q such that $M \oplus P \cong Q$;

3. *quasi-permutation* if there is an exact sequence $0 \to M \to P \to Q \to 0$ of $\mathbb{Z}[G]$-lattices in which P and Q are permutation lattices [7, §1];

These concepts are standard; see [5, §1].

Lemma 2.2. *Let H be a finite group and $K \trianglelefteq H$ a normal subgroup, and let $\mathbb{Z}[H/K]$ be the permutation $\mathbb{Z}[H]$-lattice (Definition 2.1). The character of the rational representation $\mathbb{Q}[H/K]$ is*

$$\chi_{H/K}(g) = \big|\{xK \in H/K : gxK = xK\}\big| = \begin{cases} [H:K], & g \in K, \\ 0, & g \notin K. \end{cases}$$

This lemma is used in the proof of Lemma 3.8 below, where the character of the permutation module $\mathbb{Q}[H]$ is evaluated with $K = \{1\}$.

The next definitions are those of Colliot-Thélène and Sansuc [5, §1].

Definition 2.3. Let G be a finite group and let M be a $\mathbb{Z}[G]$-lattice. Then M is *flasque* if $\hat{\mathrm{H}}^{-1}(K, M) = 0$ for every subgroup $K \leq G$, and *coflasque* if $\mathrm{H}^1(K, M) = 0$ for every subgroup $K \leq G$.

Definition 2.4. Two $\mathbb{Z}[G]$-lattices M and N are *similar* if $M \oplus S_1 \cong N \oplus S_2$ for some permutation lattices S_1 and S_2. This is an equivalence relation; the similarity class of M is written $[M]$. A $\mathbb{Z}[G]$-lattice is *invertible* if it is a direct summand of a permutation lattice [5, §1].

Proposition 2.5. *Every $\mathbb{Z}[G]$-lattice M admits a* flasque resolution

$$0 \longrightarrow M \longrightarrow P \longrightarrow F \longrightarrow 0$$

with P permutation and F flasque [5, §1, Lemme 3]. The similarity class $[F]$ depends only on M [5, §1, Lemme 5]; it is written $\rho(M)$ and called the flasque class *of M.*

Lemma 2.6. *Let G be a finite group. Let M be a $\mathbb{Z}[G]$-lattice which is a direct summand of a permutation lattice. Then M is coflasque.*

Proof. Let $M \oplus M' \cong P$ with P permutation. Cohomology is additive, so $\mathrm{H}^1(G, M)$ is a direct summand of $\mathrm{H}^1(G, P)$. It therefore suffices to show $\mathrm{H}^1(G, P) = 0$ for P permutation.

Write $P \cong \bigoplus_i \mathbb{Z}[G/K_i]$ with $K_i \leq G$. By Shapiro's Lemma [16, Theorem 6.3.2], $\mathbb{Z}[G/K_i] \cong \mathrm{Ind}_{K_i}^G(\mathbb{Z})$ and hence $\mathrm{H}^1(G, \mathbb{Z}[G/K_i]) \cong \mathrm{H}^1(K_i, \mathbb{Z})$. Since K_i acts trivially on $\mathbb{Z}$, $\mathrm{H}^1(K_i, \mathbb{Z}) \cong \mathrm{Hom}(K_i, \mathbb{Z}) = 0$. Therefore $\mathrm{H}^1(G, P) = 0$. □

Theorem 2.7. *Let G be a finite group and M a $\mathbb{Z}[G]$-lattice on which G acts faithfully. Let F be a field of characteristic* 0*. Here $F(M)$ and $F(M)^G$ are as in the Introduction, and G acts on $F(M)$ through the $\mathbb{Z}[G]$-module structure of M. If $F(M)^G$ is* stably rational *over F, then M is a quasi-permutation $\mathbb{Z}[G]$-lattice (Definition 2.1).*

In the literature this criterion is stated for the *twisted* invariant field $Q(L/K, M) = L(M)^G$ of [14, Theorem 1.3(b)] (equivalently, in the language of quasi-permutation lattices, [8, Theorem 1.6]; see also [7, §1]), where L/K is Galois with group G and G acts on the coefficient field as well; the field $F(M)^G$ above is the field $F(M, G)$ of [14, p. 221], and for it the criterion is an assumption rather than a quotation of a proved result. **We use this criterion as a hypothesis**; it is not proved in this paper.

In the sequel we apply Theorem 2.7 in the contrapositive direction, the criterion being used as a hypothesis throughout: if $M|_H$ is *not* quasi-permutation, then $F(M|_H)^H$, and therefore $Z_H(F, 9)$, is *not* stably rational. The cohomological computation of §4 proves that $\mathrm{H}^2(H, M|_H) \cong C_9$ while permutation lattices have H^2-exponent dividing 3; hence $M|_H$ is not quasi- permutation (Lemma 4.7), and the criterion yields non-stable-rationality (Corollary 4.8). The same strategy generalizes to $C_p \times C_p$ in §5.

3 The Procesi Lattice

We give a self-contained construction of the Procesi $\mathbb{Z}[S_9]$-lattice M and the Procesi isomorphism, keeping only what is needed for the rationality computations. An accessible modern exposition is in [1, §5] and [9]; the original reference is [12].

3.1 Construction over S_9

Let $n = 9$ and let $\Omega = S_9/S_8$ be the set of left cosets of S_8 in S_9. Then $|\Omega| = 9$ and S_9 acts on Ω by left multiplication. Define the permutation $\mathbb{Z}[S_9]$-module

$$U = \mathbb{Z}[\Omega] \cong \bigoplus_{\omega \in \Omega} \mathbb{Z} \cdot e_\omega,$$

where $\{e_\omega\}_{\omega \in \Omega}$ is the standard basis and $g \cdot e_\omega = e_{g\omega}$ for $g \in S_9$. Consider the $\mathbb{Z}[S_9]$-module map

$$\varphi : U \otimes_{\mathbb{Z}} U \longrightarrow \mathbb{Z}[\Omega], \qquad \varphi(e_\alpha \otimes e_\beta) = e_\alpha - e_\beta,$$

extended $\mathbb{Z}$-linearly. The map φ is S_9-equivariant:

$$\varphi\big(g\cdot(e_\alpha\otimes e_\beta)\big)=\varphi(e_{g\alpha}\otimes e_{g\beta})=e_{g\alpha}-e_{g\beta}=g\cdot(e_\alpha-e_\beta)=g\cdot\varphi(e_\alpha\otimes e_\beta).$$

The image of φ is the *augmentation submodule*

$$I[\Omega]=\ker\Big(\varepsilon:\mathbb{Z}[\Omega]\twoheadrightarrow\mathbb{Z},\ \varepsilon\big(\sum a_\omega e_\omega\big)=\sum a_\omega\Big).$$

Indeed, every difference $e_\alpha-e_\beta$ has augmentation zero, and conversely any element $\sum a_\omega e_\omega$ with $\sum a_\omega=0$ is a $\mathbb{Z}$-linear combination of such differences (take $\{e_\alpha-e_{\omega_0}:\alpha\neq\omega_0\}$ for a fixed ω_0). Moreover, φ is surjective onto $I[\Omega]$.

Definition 3.1. The *Procesi $\mathbb{Z}[S_9]$-lattice* is $M=\ker(\varphi)$.

By construction we have the short exact sequence of $\mathbb{Z}[S_9]$-modules

$$0\longrightarrow M\longrightarrow U\otimes_{\mathbb{Z}}U\xrightarrow{\ \varphi\ }I[\Omega]\longrightarrow 0. \tag{1}$$

The module M is $\mathbb{Z}$-free: it is a $\mathbb{Z}$-submodule of the free $\mathbb{Z}$-module $U\otimes_{\mathbb{Z}}U$, and every submodule of a free $\mathbb{Z}$-module is free (since $\mathbb{Z}$ is a principal ideal domain) [4, Ch.VII, §3, Th.1, Cor.2]. The ranks are

$$\operatorname{rk}_{\mathbb{Z}}U=9,\qquad \operatorname{rk}_{\mathbb{Z}}(U\otimes U)=81,\qquad \operatorname{rk}_{\mathbb{Z}}I[\Omega]=8,\qquad \operatorname{rk}_{\mathbb{Z}}M=73.$$

Proposition 3.2. *[12]; see also [1, §5] and [9]. Let F be a field of characteristic 0, with M and U as above, and let $Z_{S_9}(F,9)$ be the center of the generic matrix algebra of degree 9 over F.*

1. *The center satisfies $Z_{S_9}(F,9)\cong F(M\oplus U)^{S_9}$, where $F(M\oplus U)^{S_9}$ is the invariant field of the purely transcendental extension $F(M\oplus U)$ under the natural S_9-action induced by the $\mathbb{Z}[S_9]$-module structure of $M\oplus U$.*

2. *For every subgroup $H\subseteq S_9$, taking H-invariants gives $Z_H(F,9)=F((M\oplus U)|_H)^H$.*

3. *If H acts faithfully on $M|_H$, then $Z_H(F,9)$ has transcendence degree $82=n^2+1$ over F and is* rational *over $F(M|_H)^H$ [10, Proposition 1.4].*

The S_9 identification of this proposition, and the rationality statement of its third part for $H=C_3\times C_3$, are completely formalized in Lean in the accompanying code package; the structure the statement speaks about is constructed there, so the statement is not vacuous, and the identification requires no input beyond characteristic 0.

3.2 Restriction to $H = C_3 \times C_3$

Now fix a transitive embedding $H = C_3 \times C_3 \hookrightarrow S_9$, i.e., an embedding whose image acts transitively on $\Omega = S_9/S_8$. Since $|H| = 9 = |\Omega|$, the action is moreover regular (i.e., free and transitive, called *simply transitive* in [3, ch.I, §5, no.6, def.7]): choosing a basepoint $\omega_0 \in \Omega$, the map $H \to \Omega$, $h \mapsto h \cdot \omega_0$, is an H-equivariant bijection. Consequently we obtain an explicit isomorphism of $\mathbb{Z}[H]$-modules

$$U|_H \xrightarrow{\cong} \mathbb{Z}[H], \qquad e_{h\cdot\omega_0} \longmapsto h \quad (h \in H). \tag{2}$$

Proposition 3.3. *The isomorphism* (2) *is independent of the choice of* ω_0 *up to a* $\mathbb{Z}[H]$*-module automorphism of* $\mathbb{Z}[H]$*. Consequently,* $\mathrm{Ext}^1_{\mathbb{Z}[H]}(-, M|_H)$*,* $\mathrm{H}^2(H, M|_H)$*, and the isomorphism class of* $M|_H$ *are independent of this choice.*

Proof. Let $\omega_0' = g \cdot \omega_0$ be another basepoint. The isomorphism induced by ω_0' sends $e_{h\cdot\omega_0'} = e_{hg\cdot\omega_0} \longmapsto h$, while the original sends $e_{hg\cdot\omega_0} \longmapsto hg$; hence the two isomorphisms differ by right multiplication by g^{-1} on $\mathbb{Z}[H]$. Right multiplication by a fixed element of H is a $\mathbb{Z}[H]$-module automorphism of the left regular module. Therefore the induced map on cohomology and Ext^1 is the identity, and the isomorphism class of $M|_H$ is unchanged. □

Proposition 3.4. *Let* $\iota_1, \iota_2 : H \hookrightarrow S_9$ *be two transitive embeddings. Then* $M|_{\iota_1(H)} \cong M|_{\iota_2(H)}$ *as* $\mathbb{Z}[H]$*-modules.*

Proof. Both ι_1 and ι_2 make Ω into a regular H-set (since $|H| = 9 = |\Omega|$ and each action is transitive). Any two regular H-sets of the same cardinality are H-equivariantly isomorphic: fix basepoints $\omega_1, \omega_2 \in \Omega$, and define a bijection $\Omega \to \Omega$ by $h \cdot \omega_1 \longmapsto h \cdot \omega_2$. This bijection is an element $\sigma \in S_9$, and by construction $\sigma\, \iota_1(h)\, \sigma^{-1} = \iota_2(h)$ for all $h \in H$. Conjugating the embedding by σ replaces the H-action on $U \otimes U$ with its conjugate under the S_9-action, which induces an isomorphism $M|_{\iota_1(H)} \cong M|_{\iota_2(H)}$ of $\mathbb{Z}[H]$-modules via the identification of H with $\iota_1(H)$ and $\iota_2(H)$ followed by the conjugation $\iota_2^{-1} \circ \iota_1$. □

Proposition 3.5. *There is an isomorphism of* $\mathbb{Z}[H]$*-modules*

$$\mathbb{Z}[H] \otimes_{\mathbb{Z}} \mathbb{Z}[H] \;\cong\; \mathbb{Z}[H]^{\oplus 9},$$

where the left side carries the diagonal action $h\cdot(x\otimes y) = hx \otimes hy$ *and the right side carries the action* $h \cdot (z_b)_{b\in H} = (h z_{h^{-1}b'})_{b'\in H}$*, that is, left multiplication composed with the coordinate permutation* $b \mapsto hb$*.*

Proof. We use the group basis $\{h : h \in H\}$ of $\mathbb{Z}[H]$, so that $\mathbb{Z}[H] \otimes_{\mathbb{Z}} \mathbb{Z}[H]$ carries the diagonal action $h \cdot (x \otimes y) = hx \otimes hy$; we write $y = \sum_{a \in H} y_a\, a$ with $y_a \in \mathbb{Z}$. Define

$$\Phi : \mathbb{Z}[H] \otimes_{\mathbb{Z}} \mathbb{Z}[H] \longrightarrow \mathbb{Z}[H]^{\oplus 9}, \qquad \Phi(x \otimes y) = (\, y_a\, x\,)_{a \in H},$$

so that the a-th coordinate of $\Phi(x \otimes y)$ is $y_a x$. On a basis vector $f \otimes g$ this reads $\Phi(f \otimes g) = f$ in the g-th coordinate; hence

$$\Psi\big((z_a)_{a \in H}\big) \;=\; \sum_{a \in H} z_a \otimes a,$$

viewed as a $\mathbb{Z}$-linear map $\Psi : \mathbb{Z}[H]^{\oplus 9} \to \mathbb{Z}[H] \otimes_{\mathbb{Z}} \mathbb{Z}[H]$, is inverse to Φ: both composites are the identity on basis vectors (on the tensor side $\Phi(z_b \otimes b) = z_b$ in the b-th coordinate, and $\Psi(\Phi(x \otimes y)) = \sum_a x \otimes (y_a a) = x \otimes y$). Finally $(hy)_a = y_{h^{-1}a}$ for all $a \in H$, so

$$\Phi\big(h \cdot (x \otimes y)\big) = \Phi(hx \otimes hy) = (\, y_{h^{-1}a}\, hx\,)_{a \in H} = h \cdot (\, y_a\, x\,)_{a \in H} = h \cdot \Phi(x \otimes y)$$

for the action on the right side, and Φ is an isomorphism of $\mathbb{Z}[H]$-modules. $\square$

Under the identification (2) and Proposition 3.5, the restriction of the map φ to H yields the following.

Proposition 3.6. *There is an exact sequence of $\mathbb{Z}[H]$-modules*

$$0 \longrightarrow M|_H \longrightarrow \mathbb{Z}[H]^{\oplus 9} \xrightarrow{\pi|_H} I[H] \longrightarrow 0, \tag{3}$$

where $I[H] = \ker(\varepsilon : \mathbb{Z}[H] \twoheadrightarrow \mathbb{Z})$, $\varepsilon(\sum a_h h) = \sum a_h$, is the augmentation ideal of H. All modules in the sequence are finitely generated and $\mathbb{Z}$-free, and $\mathrm{rk}_{\mathbb{Z}}\, M|_H = 73$.

Proof. Exactness. Under the identifications, the map φ becomes a $\mathbb{Z}[H]$-module map $\pi|_H : \mathbb{Z}[H]^{\oplus 9} \longrightarrow I[H]$. Each standard basis element of $\mathbb{Z}[H]^{\oplus 9}$ maps to a difference $h - h'$ in $\mathbb{Z}[H]$; these differences generate the $\mathbb{Z}$-module $I[H]$, and $I[H]$ is generated as a $\mathbb{Z}[H]$-module by any single element $h - 1$ with $h \neq 1$; hence $\pi|_H$ is surjective. Define $M|_H = \ker(\pi|_H)$, the restriction of the Procesi lattice to H.

Freeness. All modules in the sequence are finitely generated and $\mathbb{Z}$-free: $\mathbb{Z}[H]^{\oplus 9}$ is $\mathbb{Z}$-free with basis the group elements in each coordinate; $I[H]$ is a $\mathbb{Z}$-submodule of $\mathbb{Z}[H]$, hence free [4, Ch.VII, §3, Th.1, Cor.2]; $M|_H = \ker(\pi|_H)$ is a $\mathbb{Z}$-submodule of $\mathbb{Z}[H]^{\oplus 9}$, hence free.

Rank. $\mathrm{rk}_{\mathbb{Z}}\, M|_H = \mathrm{rk}_{\mathbb{Z}}\, \mathbb{Z}[H]^{\oplus 9} - \mathrm{rk}_{\mathbb{Z}}\, I[H] = 9 \cdot 9 - 8 = 73$, matching the rank of M over S_9. $\square$

Proposition 3.7.

$$M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \;\cong\; 8 \cdot \mathbb{Q}[H] \;\oplus\; \mathbb{Q},$$

where $\mathbb{Q}$ denotes the one-dimensional trivial $\mathbb{Q}[H]$-module.

Proof. Tensor the exact sequence of Proposition 3.6 with $\mathbb{Q}$ (flat over $\mathbb{Z}$, [2, Ch.II, §2, no.4, Th.1]):

$$0 \longrightarrow M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathbb{Q}[H]^{\oplus 9} \longrightarrow I[H] \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow 0.$$

The augmentation sequence

$$0 \longrightarrow I[H] \longrightarrow \mathbb{Z}[H] \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0 \tag{4}$$

remains exact after tensoring with $\mathbb{Q}$ (flat over $\mathbb{Z}$, [2, Ch.II, §2, no.4, Th.1]), giving

$$0 \longrightarrow I[H] \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathbb{Q}[H] \xrightarrow{\varepsilon} \mathbb{Q} \longrightarrow 0.$$

This sequence splits since $|H| = 9$ is invertible in $\mathbb{Q}$ (Maschke's Theorem [6, (3.14)]; the splitting is given by the idempotent $e_0 = \frac{1}{|H|}\sum_{h\in H} h$).

We now compute $\ker \pi|_H$ explicitly. Write $z_b = \sum_{x\in H}(z_b)_x\, x$ with $(z_b)_x \in \mathbb{Q}$ for an element z of $\mathbb{Q}[H]^{\oplus 9}$, and define

$$\Theta(z)_t \;=\; \sum_{x\in H} (z_{tx})_x\, x, \qquad t \in H.$$

The coefficient identity $(\Theta(z)_t)_y = (z_{ty})_y$ is equivalent to $(z_b)_y = ((\Theta z)_{by^{-1}})_y$; hence $\Theta^{-1}(W)_b = \sum_{x\in H}(W_{bx^{-1}})_x\, x$ is inverse to Θ, with

$$(\Theta^{-1}(\Theta(z))_b)_y = (\Theta(z)_{by^{-1}})_y = (z_b)_y, \quad (\Theta(\Theta^{-1}(W))_t)_y = (\Theta^{-1}(W)_{ty})_y = (W_t)_y.$$

Moreover

$$\big(\Theta(h\cdot z)_t\big)_y = (z_{h^{-1}ty})_{h^{-1}y} = \big(h\cdot \Theta(z)_t\big)_y,$$

using commutativity of H for $h^{-1}ty = t\,h^{-1}y$; hence Θ is an isomorphism from $\mathbb{Q}[H]^{\oplus 9}$ with the action $h\cdot(z_b) = (h\, z_{h^{-1}b})$ to $\mathbb{Q}[H]^{\oplus 9}$ with the coordinatewise action $h\cdot(z_b) = (h\, z_b)$. Under this identification the map $\pi|_H$ corresponds to

$$\pi_{\mathrm{std}}(z) \;=\; \sum_{b\in H} z_b\,(1-b)\,:$$

indeed Θ^{-1} fixes the standard coordinate vectors u_b (the vector with the unit 1 in coordinate b), and $\pi|_H(u_b) = 1-b$, since under (2) and Proposition 3.5 the vector u_b corresponds to $e_{\omega_0}\otimes e_{b\cdot\omega_0}$, whose image under φ is the difference $e_{\omega_0} - e_{b\cdot\omega_0}$. Both sides are $\mathbb{Q}[H]$-linear for the coordinatewise action (for π_{std}

this follows since $\pi|_H$ is $\mathbb{Z}[H]$-linear and Θ is an isomorphism for the two structures), and they agree on the $\mathbb{Q}[H]$-basis $\{u_b\}_{b\in H}$; hence they coincide. Consequently, identifying $M|_H\otimes_{\mathbb{Z}}\mathbb{Q}$ with the kernel of the $\mathbb{Q}$-linear extension of $\pi|_H$ to $\mathbb{Q}[H]^{\oplus 9}$, the map Θ restricts to an isomorphism $M|_H\otimes_{\mathbb{Z}}\mathbb{Q}\cong E=\ker\pi_{\mathrm{std}}$, a $\mathbb{Q}[H]$-submodule of $\mathbb{Q}[H]^{\oplus 9}$ for the coordinatewise action. It remains to decompose E.

Let

$$\pi_1\colon \mathbb{Q}[H]^{\oplus 9}\longrightarrow \mathbb{Q}[H]^{\oplus 8}\times\mathbb{Q},\qquad z\longmapsto \big((z_b)_{b\neq 1},\ \varepsilon(z_1)\big),$$

$$\sigma\colon \mathbb{Q}[H]^{\oplus 8}\longrightarrow \mathbb{Q}[H],\qquad y\longmapsto \sum_{b\neq 1} y_b\,(1-b),$$

which are $\mathbb{Q}[H]$-linear for the coordinatewise action ($\varepsilon(hz_1)=\varepsilon(z_1)$). Put $K=\ker\sigma$ and $\widetilde{I}=\{z : z_b=0 \text{ for } b\neq 1,\ z_1\in I[H]\otimes\mathbb{Q}\}$. Since $\pi_{\mathrm{std}}(z)=\sigma((z_b)_{b\neq 1})$, we have $E=\pi_1^{-1}(K\times\mathbb{Q})$, and π_1 restricts to an exact sequence of $\mathbb{Q}[H]$-modules

$$0\longrightarrow \widetilde{I}\longrightarrow E\xrightarrow{\pi_1} K\times\mathbb{Q}\longrightarrow 0.$$

Indeed π_1 maps E onto $K\times\mathbb{Q}$: for $y\in K$ and $c\in\mathbb{Q}$ the element with coordinates y on $b\neq 1$ and $c\cdot 1$ on $b=1$ lies in E and is mapped to (y,c). Its kernel on E consists of the z with $z_b=0$ for $b\neq 1$ and $\varepsilon(z_1)=0$, that is, of $\widetilde{I}$. Moreover the elements $1-b$, $b\neq 1$, form a $\mathbb{Q}$-basis of $I[H]\otimes\mathbb{Q}$ (they span, as $I[H]$ is generated by differences, and are linearly independent: the coefficient of $y\neq 1$ in $\sum_{b\neq 1}c_b\,(1-b)$ is $-c_y$); hence σ is surjective and

$$0\longrightarrow K\longrightarrow \mathbb{Q}[H]^{\oplus 8}\xrightarrow{\sigma} I[H]\otimes\mathbb{Q}\longrightarrow 0$$

is exact. Both exact sequences split, because $\mathbb{Q}[H]$ is semisimple by Maschke [6, (3.14)] and all modules involved are finitely generated. Finally $\widetilde{I}\cong I[H]\otimes\mathbb{Q}$ as $\mathbb{Q}[H]$-modules via $z\mapsto z_1$ (with inverse the coordinate inclusion). Combining the splittings,

$$E\ \cong\ \widetilde{I}\oplus K\oplus\mathbb{Q}\ \cong\ (I[H]\otimes\mathbb{Q})\oplus K\oplus\mathbb{Q}\ \cong\ \mathbb{Q}[H]^{\oplus 8}\oplus\mathbb{Q},$$

and, recalling that $M|_H\otimes_{\mathbb{Z}}\mathbb{Q}\cong E$ under Θ, this proves that $M|_H\otimes_{\mathbb{Z}}\mathbb{Q}\cong 8\cdot\mathbb{Q}[H]\oplus\mathbb{Q}$ as $\mathbb{Q}[H]$-modules. □

From Proposition 3.7 it follows that if $M|_H$ were a permutation $\mathbb{Z}[H]$-lattice, then $M|_H\cong 8\cdot\mathbb{Z}[H]\oplus\mathbb{Z}$ (Lemma 3.8 below).

Lemma 3.8. *Let P be a permutation $\mathbb{Z}[H]$-lattice satisfying $P\otimes_{\mathbb{Z}}\mathbb{Q}\cong 8\cdot\mathbb{Q}[H]\oplus\mathbb{Q}$. Then $P\cong 8\cdot\mathbb{Z}[H]\oplus\mathbb{Z}$.*

Proof. A permutation $\mathbb{Z}[H]$-lattice is, equivalently, the permutation module $\mathbb{Z}[X]$ of a finite H-set X (Definition 2.1). By flatness of $\mathbb{Q}$ over $\mathbb{Z}$ and Proposition 3.7,

$$P \otimes_{\mathbb{Z}} \mathbb{Q} \;\cong\; 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$$

as $\mathbb{Q}[H]$-modules. For $g \in H$, the trace of g on $8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$ equals $8 \cdot \#\{h \in H : gh = h\} + 1$, where $\#\{h : gh = h\}$ is 9 for $g = 1$ and 0 otherwise (Lemma 2.2 with $K = \{1\}$); this is 73 for $g = 1$ and 1 for $g \neq 1$. On the other hand, the trace of g on $P \otimes \mathbb{Q}$ equals the trace of g on the permutation module $\mathbb{Z}[X]$, which is the number of fixed points of g on X. Comparing, $|X| = 73$ and g has exactly one fixed point in X for every $g \neq 1$.

Let $g \neq 1$ and let x_g be its unique fixed point in X. For every $h \in H$ the point hx_g is again fixed by g, since $g(hx_g) = h(gx_g) = hx_g$ by commutativity of H; uniqueness forces $hx_g = x_g$. Thus x_g is fixed by all of H. Fix $g_0 \neq 1$, which exists as $|H| = 9$. Its fixed point x_{g_0} is then the unique H-fixed point of X: an H-fixed point is fixed by g_0, hence equals x_{g_0}. Write $x_0 = x_{g_0}$.

If $x \in X \setminus \{x_0\}$ were fixed by some $g \neq 1$, then $x = x_g = x_0$, a contradiction. Hence H acts freely on $X \setminus \{x_0\}$, so this set is a disjoint union of orbits of cardinality $|H| = 9$; as $|X \setminus \{x_0\}| = 72 = 8 \cdot 9$, it is the disjoint union of eight free orbits. Therefore $X \cong \{x_0\} \sqcup H^{\sqcup 8}$ as H-sets, and $P \cong \mathbb{Z}[X] \cong \mathbb{Z} \oplus 8 \cdot \mathbb{Z}[H]$. □

4 Cohomological Obstruction to Quasi-Permutation

In this section we compute the second integral cohomology group $\mathrm{H}^2(H, M|_H)$ and prove that its exponent obstructs quasi-permutation.

Notation. Throughout this section we write $\mathrm{H}^q(G, M)$ for the q-th integral group cohomology of a G-module M. The upright H is reserved for the cohomology functor, while the italic H denotes the group $H = C_3 \times C_3$.

4.1 Cohomology Computations

Lemma 4.1. *The group* $\mathrm{H}^1(H, I[H])$ *is cyclic of order* 9.

Proof. Apply group cohomology $\mathrm{H}^*(H, -)$ to the augmentation exact sequence (4). The associated long exact cohomology sequence reads

$$0 \longrightarrow \mathrm{H}^0(H, I[H]) \longrightarrow \mathrm{H}^0(H, \mathbb{Z}[H]) \longrightarrow \mathrm{H}^0(H, \mathbb{Z}) \longrightarrow \mathrm{H}^1(H, I[H]) \longrightarrow \mathrm{H}^1(H, \mathbb{Z}[H]) \longrightarrow \cdots .$$

We analyse the low-degree terms:

- $\mathrm{H}^0(H, \mathbb{Z}[H]) = \mathbb{Z}[H]^H = \mathbb{Z} \cdot N_H$, where $N_H = \sum_{h \in H} h$ is the norm element.
- $\mathrm{H}^0(H, \mathbb{Z}) = \mathbb{Z}^H = \mathbb{Z}$.
- The map $\mathrm{H}^0(H, \mathbb{Z}[H]) \to \mathrm{H}^0(H, \mathbb{Z})$ sends $N_H \mapsto \varepsilon(N_H) = \sum_{h \in H} 1 = |H| = 9$.
- $\mathrm{H}^1(H, \mathbb{Z}[H]) = 0$: $\mathbb{Z}[H] \cong \mathrm{Ind}_1^H \mathbb{Z}$ as $\mathbb{Z}[H]$-modules, and Shapiro's Lemma [16, Theorem 6.3.2] gives $\mathrm{H}^1(H, \mathrm{Ind}_1^H \mathbb{Z}) \cong \mathrm{H}^1(\{1\}, \mathbb{Z}) = 0$.

From the exactness at $\mathrm{H}^0(H, \mathbb{Z})$ we obtain $\mathrm{coker}(\mathbb{Z} \cdot N_H \xrightarrow{\varepsilon} \mathbb{Z}) = \mathbb{Z}/9\mathbb{Z}$, and exactness at $\mathrm{H}^1(H, I[H])$ with the vanishing of $\mathrm{H}^1(H, \mathbb{Z}[H])$ yields the short exact sequence

$$0 \longrightarrow \mathbb{Z}/9\mathbb{Z} \longrightarrow \mathrm{H}^1(H, I[H]) \longrightarrow 0,$$

hence $\mathrm{H}^1(H, I[H]) \cong \mathbb{Z}/9\mathbb{Z} \cong C_9$. □

Lemma 4.2. *For the Procesi restriction $M|_H$, $\mathrm{H}^2(H, M|_H) \cong C_9$.*

Proof. Apply $\mathrm{H}^*(H, -)$ to the exact sequence of Proposition 3.6:

$$0 \longrightarrow M|_H \longrightarrow \mathbb{Z}[H]^{\oplus 9} \longrightarrow I[H] \longrightarrow 0.$$

The associated long exact cohomology sequence gives

$$\begin{aligned} \cdots \longrightarrow \mathrm{H}^1(H, \mathbb{Z}[H]^{\oplus 9}) &\longrightarrow \mathrm{H}^1(H, I[H]) \\ \longrightarrow \mathrm{H}^2(H, M|_H) &\longrightarrow \mathrm{H}^2(H, \mathbb{Z}[H]^{\oplus 9}) \longrightarrow \cdots . \end{aligned}$$

By Shapiro's Lemma [16, Theorem 6.3.2], $\mathrm{H}^i(H, \mathbb{Z}[H]) \cong \mathrm{H}^i(\{1\}, \mathbb{Z}) = 0$ for all $i > 0$. Hence $\mathrm{H}^1(H, \mathbb{Z}[H]^{\oplus 9}) = \mathrm{H}^2(H, \mathbb{Z}[H]^{\oplus 9}) = 0$, and the long exact sequence collapses to the isomorphism

$$\mathrm{H}^2(H, M|_H) \;\cong\; \mathrm{H}^1(H, I[H]).$$

By Lemma 4.1, $\mathrm{H}^1(H, I[H]) \cong C_9$. □

Lemma 4.3. *With the trivial H-action on $\mathbb{Z}$, $\mathrm{H}^2(H, \mathbb{Z}) \cong C_3 \times C_3$.*

Proof. Apply $\mathrm{H}^*(H, -)$ to the short exact sequence of trivial H-modules

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0.$$

The associated long exact sequence reads

$$\begin{aligned} 0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} & \\ \longrightarrow \mathrm{H}^1(H, \mathbb{Z}) \longrightarrow \mathrm{H}^1(H, \mathbb{Q}) &\longrightarrow \mathrm{H}^1(H, \mathbb{Q}/\mathbb{Z}) \\ \longrightarrow \mathrm{H}^2(H, \mathbb{Z}) \longrightarrow \mathrm{H}^2(H, \mathbb{Q}) &\longrightarrow \cdots . \end{aligned}$$

Since $\mathbb{Q}$ is uniquely divisible, multiplication by 3 is an automorphism of $\mathbb{Q}$; hence $\mathrm{H}^i(H, \mathbb{Q}) = 0$ for all $i > 0$ [11, Proposition 1.6.2(c)]. Substituting $\mathrm{H}^1(H, \mathbb{Q}) = \mathrm{H}^2(H, \mathbb{Q}) = 0$ yields $\mathrm{H}^2(H, \mathbb{Z}) \cong \mathrm{H}^1(H, \mathbb{Q}/\mathbb{Z})$. Since $\mathbb{Q}/\mathbb{Z}$ carries the trivial H-action, $\mathrm{H}^1(H, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(H, \mathbb{Q}/\mathbb{Z})$ [16, Corollary 6.4.6]. Now $\mathrm{Hom}(H, \mathbb{Q}/\mathbb{Z}) \cong C_3 \times C_3$ because $\mathrm{Hom}(C_3, \mathbb{Q}/\mathbb{Z}) \cong C_3$ ($\mathbb{Q}/\mathbb{Z}$ has a unique subgroup of order 3) and $\mathrm{Hom}(-, \mathbb{Q}/\mathbb{Z})$ preserves finite direct sums [3, Ch.II, §1, no.6, Prop.6, Cor.1]. Hence $\mathrm{H}^2(H, \mathbb{Z}) \cong C_3 \times C_3$. □

4.2 Exponent of H^2 for Permutation Lattices

Proposition 4.4. *Let P be a permutation $\mathbb{Z}[H]$-lattice. Then every element of $\mathrm{H}^2(H, P)$ has order dividing 3; equivalently, $\exp(\mathrm{H}^2(H, P)) \mid 3$.*

Proof. Write $P \cong \bigoplus_i \mathbb{Z}[H/K_i]$ for subgroups $K_i \leq H$. Since $\mathrm{H}^2(H, -)$ is additive and therefore preserves finite direct sums [16, Proposition 3.3.4], $\mathrm{H}^2(H, P) \cong \bigoplus_i \mathrm{H}^2(H, \mathbb{Z}[H/K_i])$. By Shapiro's Lemma [16, Theorem 6.3.2], $\mathbb{Z}[H/K_i] \cong \mathrm{Ind}_{K_i}^H(\mathbb{Z})$, hence $\mathrm{H}^2(H, \mathbb{Z}[H/K_i]) \cong \mathrm{H}^2(K_i, \mathbb{Z})$, where $\mathbb{Z}$ carries the trivial K_i-action. For each subgroup $K \leq H$ we compute $\mathrm{H}^2(K, \mathbb{Z})$:

- $K = \{1\}$: $\mathrm{H}^2(\{1\}, \mathbb{Z}) = 0$.
- $K \cong C_3$ (four subgroups of order 3): $\mathrm{H}^2(C_3, \mathbb{Z}) \cong \mathrm{Hom}(C_3, \mathbb{Q}/\mathbb{Z}) \cong C_3$.
- $K = H = C_3 \times C_3$: $\mathrm{H}^2(H, \mathbb{Z}) \cong C_3 \times C_3$ (Lemma 4.3).

In every case $\exp(\mathrm{H}^2(K, \mathbb{Z})) \mid 3$. Hence $\exp(\mathrm{H}^2(H, P)) \mid 3$. □

Proposition 4.5. *If a $\mathbb{Z}[H]$-lattice L is stably permutation, then*

$$\exp(\mathrm{H}^2(H, L)) \mid 3.$$

Proof. Write $L \oplus P \cong Q$ with P, Q permutation. By additivity [16, Proposition 3.3.4], $\mathrm{H}^2(H, L) \oplus \mathrm{H}^2(H, P) \cong \mathrm{H}^2(H, Q)$. Both $\mathrm{H}^2(H, P)$ and $\mathrm{H}^2(H, Q)$ have exponent dividing 3 by Proposition 4.4. For any $x \in \mathrm{H}^2(H, L)$, the element $(x, 0) \in \mathrm{H}^2(H, Q)$ satisfies $3(x, 0) = 0$; hence $3x = 0$. Therefore $\exp(\mathrm{H}^2(H, L)) \mid 3$. □

4.3 The Obstruction

Theorem 4.6. *The restricted Procesi lattice $M|_H$ is not a stably permutation $\mathbb{Z}[H]$-lattice.*

Proof. By Lemma 4.2, $\mathrm{H}^2(H, M|_H) \cong C_9$, a group of exponent 9. If $M|_H$ were stably permutation, Proposition 4.5 would force $\exp(\mathrm{H}^2(H, M|_H)) \mid 3$. Contradiction. □

Lemma 4.7. *The lattice $M|_H$ is not quasi-permutation (Definition 2.1).*

Proof. Suppose there are permutation lattices P, Q and a short exact sequence $0 \to M|_H \to P \to Q \to 0$. The associated long exact cohomology sequence gives

$$\mathrm{H}^1(H, Q) \longrightarrow \mathrm{H}^2(H, M|_H) \longrightarrow \mathrm{H}^2(H, P).$$

The group $\mathrm{H}^1(H, Q)$ vanishes: by Shapiro's lemma [16, Theorem 6.3.2], $\mathrm{H}^1(H, \mathbb{Z}[H/K]) \cong \mathrm{H}^1(K, \mathbb{Z}) = 0$ for every subgroup $K \leq H$. Hence $\mathrm{H}^2(H, M|_H)$ embeds into $\mathrm{H}^2(H, P)$, a group of exponent dividing 3 (Proposition 4.4). This is impossible since $\mathrm{H}^2(H, M|_H) \cong C_9$ has exponent 9 (Lemma 4.2). □

Corollary 4.8. *The field $Z_H(F, 9)$ is not stably rational over F.*

Proof. Recall that $Z_H(F, 9)$ is rational over $F(M|_H)^H$ (Proposition 3.2), the action of H on $M|_H$ being faithful (Proposition 4.9). By Lemma 4.7, $M|_H$ is not quasi-permutation. By Theorem 2.7, used as a hypothesis, stable rationality of $F(M|_H)^H$ over F forces $M|_H$ to be quasi-permutation. Hence $F(M|_H)^H$, and therefore $Z_H(F, 9)$, is not stably rational. □

Proposition 4.9. *Let $M_9(F)$ denote the algebra of 9×9 matrices over F. Let $H = C_3 \times C_3$ act on $M_9(F)$ via the regular representation $H \hookrightarrow S_9$ by permutation of matrix indices: $h \cdot e_{ij} = e_{h(i),h(j)}$ for the standard matrix units. The action of H on $M|_H$ is the restriction of the diagonal left-regular action on $\mathbb{Z}[H]^{\oplus 9}$ of Proposition 3.5:*

$$h \cdot (z_b)_{b \in H} = (h\, z_{h^{-1}b'})_{b' \in H}, \qquad h \in H,\ (z_b)_{b \in H} \in M|_H \subset \mathbb{Z}[H]^{\oplus 9},$$

that is, left multiplication composed with the coordinate permutation $b \mapsto hb$. This action is faithful.

Proof. We show that the action on $M|_H$ is faithful. Suppose that $h \in H$ acts trivially on $M|_H$. Since $M|_H = \ker(\pi|_H)$ by Proposition 3.6, the difference map

$$\delta_h(z) \;=\; h \cdot x - x, \qquad x \in \mathbb{Z}[H]^{\oplus 9} \text{ any lift of } z \in I[H],$$

is well defined on $I[H] = \mathrm{im}(\pi|_H)$: if $x - x' \in \ker(\pi|_H) = M|_H$, then $h \cdot (x - x') = x - x'$. Identify $\mathbb{Z}[H]^{\oplus 9} \cong \mathbb{Z}[H] \otimes_{\mathbb{Z}} \mathbb{Z}[H]$ with basis $\{\delta_b\}_{b \in H}$ in the second factor, as in Proposition 3.5; the diagonal action reads $h \cdot (a \otimes \delta_b) = ha \otimes \delta_{hb}$, and, as in the Second proof below,

$$\pi|_H\Big(\sum_{b \in H} z_b \otimes \delta_b\Big) = \sum_{b \in H} \big(z_b - \varepsilon(z_b)\, b\big).$$

Choose $g \in H \setminus \{1, h, h^{-1}\}$, which exists because $|H| = 9 > 3$. We compute $\delta_h(g-1)$ in two ways. First, $g \otimes \delta_1$ maps to $g - 1$ under $\pi|_H$, so

$$\delta_h(g-1) = h \cdot (g \otimes \delta_1) - g \otimes \delta_1 = hg \otimes \delta_h - g \otimes \delta_1.$$

Second, $-(1 \otimes \delta_g)$ also maps to $-\big(1 - \varepsilon(1)\, g\big) = g - 1$, so

$$\delta_h(g-1) = -h \cdot (1 \otimes \delta_g) + 1 \otimes \delta_g = 1 \otimes \delta_g - h \otimes \delta_{hg}.$$

The δ_{hg}-component of the right-hand side of the first display is 0, because $\delta_{hg} \neq \delta_h$ and $\delta_{hg} \neq \delta_1$ (as $g \neq 1$ and $g \neq h^{-1}$), whereas in the second display it is $-h \neq 0$. This contradiction shows that no $h \neq 1$ acts trivially on $M|_H$; the action is faithful. □

Second proof. Let $h \in H$, $h \neq 1$, and let $z \in \mathbb{Z}[H]^{\oplus 9}$ be the element with coordinates $z_1 = h - 1$, $z_h = 1 - h$, and $z_b = 0$ for $b \notin \{1, h\}$. Under the identifications of Proposition 3.5, the element with coordinates $(z_b)_{b \in H}$ corresponds to $\sum_{b \in H} z_b \otimes \delta_b$, where δ_b denotes the group element b of the second factor, identified with $e_{b \cdot \omega_0}$. Since $\varphi(e_\alpha \otimes e_\beta) = e_\alpha - e_\beta$, the map $\pi|_H$ sends this element to $\sum_{b \in H}(z_b - \varepsilon(z_b)\, b)$. Since $\varepsilon(h-1) = \varepsilon(1-h) = 0$, this gives $\pi|_H(z) = (h-1) + (1-h) = 0$, and $z \in M|_H$. Now act by h. By Proposition 3.5 the h^{-1}-th coordinate of z moves into the first coordinate and is multiplied by h, so $(h \cdot z)_1 = h \cdot z_{h^{-1}}$. Here $h^{-1} \neq 1$ because $h \neq 1$, and $h^{-1} \neq h$: if $h^{-1} = h$, then $h^2 = 1$, and $h = h^3(h^2)^{-1} = 1$ since $h^3 = 1$. Hence $z_{h^{-1}} = 0$, and $(h \cdot z)_1 = 0$, whereas $z_1 = h - 1 \neq 0$. Therefore $h \cdot z \neq z$, that is, h acts nontrivially on $M|_H$. □

Theorem 4.10. *Let G be a finite group and let M be a $\mathbb{Z}[G]$-lattice on which G acts faithfully. Let F be a field of characteristic 0. Then $F(M)^G$ is retract rational over F if and only if the flasque class $\rho(M)$ (Proposition 2.5) is the class of an invertible lattice (Definition 2.4).*

In the literature this criterion is stated for the *twisted* invariant field $Q(L/K, M) = L(M)^G$ of [13, Theorem 3.14(a)], where the invariant attached to M is its flasque class. As for Theorem 2.7, the field $F(M)^G$ above is the field $F(M, G)$ of [14, p. 221], and for it the criterion is an assumption rather than a quotation of a proved result.

Theorem 4.11. *Let $H = C_3 \times C_3$ and let F be a field of characteristic 0. Then $Z_H(F, 9) = F(M|_H)^H$ is* not *retract rational over F.*

Proof. By Theorem 4.10 it suffices to show that the flasque class $\rho(M|_H)$ is not invertible. We show the stronger statement that $\rho(M|_H)$ is not coflasque; then Lemma 2.6 applies.

Let P be a permutation $\mathbb{Z}[H]$-lattice containing $M|_H$ with finite index. Then P and $M|_H$ have the same rank, so Proposition 3.7 gives

$$P \otimes_{\mathbb{Z}} \mathbb{Q} \;\cong\; M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \;\cong\; 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}.$$

Write $P \cong \bigoplus_{K \le H} m_K \cdot \mathbb{Z}[H/K]$ with $m_K \ge 0$. The multiplicity of the regular representation in $P \otimes_{\mathbb{Z}} \mathbb{Q}$ equals $m_{\{1\}}$, hence $m_{\{1\}} = 8$. The multiplicity of the trivial representation equals $\sum_K m_K$, hence $\sum_K m_K = 9$. The rank of P equals $\sum_K m_K[H:K]$, hence

$$73 \;=\; \sum_K m_K[H:K] \;=\; 72 \;+\; \sum_{K \neq \{1\}} m_K[H:K].$$

Since $\sum_{K \neq \{1\}} m_K = 1$, the remaining orbit has index 1, that is, $K = H$. Therefore

$$P \;\cong\; \mathbb{Z}[H]^{\oplus 8} \oplus \mathbb{Z}.$$

By Shapiro's Lemma and Lemma 4.3,

$$\mathrm{H}^2(H, P) \;\cong\; \mathrm{H}^2(H, \mathbb{Z}) \;\cong\; C_3 \times C_3,$$

and in particular $\exp(\mathrm{H}^2(H,P)) \mid 3$ for every such P. On the other hand $\mathrm{H}^2(H, M|_H) \cong C_9$ by Theorem 4.6.

Let $E = P/M|_H$. The short exact sequence $0 \to M|_H \to P \to E \to 0$ induces an exact sequence

$$\mathrm{H}^1(H, E) \longrightarrow \mathrm{H}^2(H, M|_H) \xrightarrow{\alpha} \mathrm{H}^2(H, P).$$

The group $\mathrm{H}^2(H, M|_H)$ is cyclic of order 9, and $\mathrm{H}^2(H,P)$ has exponent dividing 3. Hence the image of α has order dividing 3, so $\ker(\alpha) \neq 0$. By exactness $\mathrm{H}^1(H,E) \neq 0$, and E is not coflasque.

Now let $0 \to M|_H \to P \to F \to 0$ be a flasque resolution of $M|_H$ (Proposition 2.5). Its middle term P is a permutation lattice containing $M|_H$ with finite index, so the preceding paragraph applies to it, and F is not coflasque. Since an invertible lattice is coflasque (Lemma 2.6), the class $\rho(M|_H) = [F]$ is not invertible. By Theorem 4.10, $F(M|_H)^H$ is not retract rational over F. □

Proposition 4.12. *The restricted Procesi lattice $M|_H$ is not a permutation $\mathbb{Z}[H]$-lattice.*

Proof. Apply $\mathrm{Hom}_{\mathbb{Z}[H]}(-, L)$ to (4). The associated long exact sequence reads

$$\begin{aligned} 0 \longrightarrow \mathrm{Hom}_{\mathbb{Z}[H]}(\mathbb{Z}, L) &\longrightarrow \mathrm{Hom}_{\mathbb{Z}[H]}(\mathbb{Z}[H], L) \longrightarrow \mathrm{Hom}_{\mathbb{Z}[H]}(I[H], L) \\ &\longrightarrow \mathrm{Ext}^1_{\mathbb{Z}[H]}(\mathbb{Z}, L) \longrightarrow \mathrm{Ext}^1_{\mathbb{Z}[H]}(\mathbb{Z}[H], L) \longrightarrow \mathrm{Ext}^1_{\mathbb{Z}[H]}(I[H], L) \\ &\longrightarrow \mathrm{Ext}^2_{\mathbb{Z}[H]}(\mathbb{Z}, L) \longrightarrow \mathrm{Ext}^2_{\mathbb{Z}[H]}(\mathbb{Z}[H], L) \longrightarrow \cdots . \end{aligned}$$

Since $\mathbb{Z}[H]$ is projective, $\mathrm{Ext}^i_{\mathbb{Z}[H]}(\mathbb{Z}[H], -) = 0$ for all $i \geq 1$. Substituting this collapses the sequence to an isomorphism $\mathrm{Ext}^1_{\mathbb{Z}[H]}(I[H], L) \cong \mathrm{Ext}^2_{\mathbb{Z}[H]}(\mathbb{Z}, L) = \mathrm{H}^2(H, L)$. Taking $L = M|_H$ and $L = 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$ yields

$$\mathrm{Ext}^1_{\mathbb{Z}[H]}(I[H], M|_H) \cong C_9, \qquad \mathrm{Ext}^1_{\mathbb{Z}[H]}(I[H], 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}) \cong C_3 \times C_3.$$

By Proposition 3.7, $M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \cong 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$. If $M|_H$ were a permutation lattice, Lemma 3.8 would then give $M|_H \cong 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$, and the two groups above would coincide, contradicting $C_9 \ncong C_3 \times C_3$. □

Second proof. By Lemma 4.2, $\mathrm{H}^2(H, M|_H) \cong C_9$, a group of exponent 9. If $M|_H$ were a permutation $\mathbb{Z}[H]$-lattice, Proposition 4.4 would force $\exp(\mathrm{H}^2(H, M|_H)) \mid 3$. Contradiction. □

Remark 4.13. Proposition 4.12 is strictly weaker than Theorem 4.6, and the latter is strictly weaker than Lemma 4.7. Indeed a permutation lattice is stably permutation (take $P = 0$) and a stably permutation lattice is quasi-permutation (take the sequence $0 \to M \to Q \to P \to 0$ associated with $M \oplus P \cong Q$), and both inclusions are strict: for $G = C_2 \times C_2$ the direct sum $\chi_1 \oplus \chi_2$ of two distinct non-trivial characters is stably permutation since $\chi_1 \oplus \chi_2 \oplus \mathbb{Z}[G/\ker(\chi_1\chi_2)] \cong \mathbb{Z}[G]$, while it is not permutation: the character of every non-zero permutation lattice $\mathbb{Z}[G/K]$ contains the trivial character, whereas $\chi_1 + \chi_2$ does not; and for $G = C_3$ the augmentation ideal $I[C_3]$ is quasi-permutation via $0 \to I[C_3] \to \mathbb{Z}[C_3] \to \mathbb{Z} \to 0$, but not stably permutation: passing to the coinvariants $- \otimes_{\mathbb{Z}[C_3]} \mathbb{Z}$ maps it onto C_3, while permutation lattices over the cyclic group C_3 have torsion-free coinvariants. Thus non-permutation does not imply non-stable permutation, and non-stable permutation does not imply non-quasi-permutation. The main proof therefore uses Lemma 4.7 together with the implication "stably rational ⇒ quasi-permutation" (Theorem 2.7).

5 Generalization to $C_p \times C_p$

We record the generalization of Corollary 4.8 to all odd primes.

Theorem 5.1. *Let p be an odd prime and let $H = C_p \times C_p \subset S_{p^2}$ act regularly (i.e., freely and transitively on $\{1, 2, \ldots, p^2\}$). Let $M|_H$ be the restriction of the Procesi $\mathbb{Z}[S_{p^2}]$-lattice to H. Let F be a field of characteristic 0, and let $Z_H(F, p^2)$ denote the center of the generic division algebra of degree p^2 over F (cf. Proposition 3.2), rational over $F(M|_H)^H$. Then:*

1. $\mathrm{H}^2(H, M|_H) \cong C_{p^2}$.

2. *For any permutation $\mathbb{Z}[H]$-lattice P,* $\exp(\mathrm{H}^2(H,P)) \mid p$.

3. *$M|_H$ is not quasi-permutation.*

4. *The field $Z_H(F,p^2)$ is neither stably rational nor rational over F.*

5. *The retract rationality of $Z_H(F,p^2)$ is settled negatively for $p=3$ (Theorem 4.11) and is left open for $p>3$.*

Proof. The Procesi restriction to H follows the same construction as in §3: the regular identification $U|_H \cong \mathbb{Z}[H]$ (Proposition 3.3) and the isomorphism $\mathbb{Z}[H]\otimes\mathbb{Z}[H] \cong \mathbb{Z}[H]^{\oplus p^2}$ (Proposition 3.5, whose proof works verbatim with 9 replaced by p^2) hold integrally for all p. The map $\pi|_H : \mathbb{Z}[H]^{\oplus p^2} \to I[H]$ is surjective because $I[H]$ is generated by $h-1$ ($h \neq 1$), and $M|_H = \ker(\pi|_H)$ is $\mathbb{Z}$-free (submodule of a free $\mathbb{Z}$-module) of $\mathbb{Z}$-rank p^4-p^2+1.

(1) The same cohomological computation as in Lemmas 4.1 and 4.2, with 9 replaced by p^2 and $H = C_p \times C_p$, gives $\mathrm{H}^2(H, M|_H) \cong C_{p^2}$.

(2) The same argument as in Proposition 4.4, with the subgroups of $C_p \times C_p$ in place of those of $C_3 \times C_3$, gives $\exp(\mathrm{H}^2(H,P)) \mid p$.

(3) The same argument as in Lemma 4.7 and Corollary 4.8, with p in place of 3, shows that $M|_H$ is not quasi-permutation and $Z_H(F,p^2)$ is not stably rational. □

Remark 5.2. The proof of Theorem 5.1 holds for all odd primes p without any dependence on $p=3$ specifically: the regular embedding $H \hookrightarrow S_{p^2}$ (independent of the choice up to conjugation; Proposition 3.4), Proposition 3.5, the cohomological vanishing (Shapiro), and the subgroup analysis of $C_p \times C_p$ are all purely formal and independent of p. The converse implication, from rationality back to stable permutation, remains open for non-cyclic p-groups; its twisted counterparts are [8, Theorem 1.6] and [14, Theorem 1.3(b)]. It is not required for any of the above conclusions.

Our results on [1, Problem 5.2] are summarized in Table 1.

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